UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 00498

LH QUANTUM MECHANICS 3

SEMESTER 1 EXAMINATIONS 2022/23

Time allowed: 1 hour 30 minutes

*Answer **two** questions. If you answer more than two questions, credit will only be given for the* best *two answers.*

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

A formula sheet will be found at the end of this question paper.

No calculator is permitted in this examination.

1. A two-level system has zero energy in its ground state, $|0\rangle$, and a positive energy ε in its excited state, $|1\rangle$. The energy basis $\{\,|n\rangle\}$ (with $n = 0,\, 1$) is assumed to be orthonormal. A certain observable B has a corresponding operator $\hat{B}$ given in this basis by

$$\hat{B} = -\frac{3}{5}\Big(\,|0\rangle\langle 0| - \,|1\rangle\langle 1|\Big) + \frac{4}{5}\Big(\,|0\rangle\langle 1| + \,|1\rangle\langle 0|\Big)\,.$$

(a) Write down a Hamiltonian of the system in basis $\{\,|n\rangle\}$. **[3]**

(b) By constructing a matrix representation of $\hat{B}$ in this basis, show that its eigenvalues are equal to ± 1. **[4]**

(c) Find the corresponding orthonormalised eigenvectors of $\hat{B}$, called $|+\rangle$ and $|-\rangle$ in Dirac representation. **[6]**

(d) The system was prepared in the excited state before a measurement of $\hat{B}$ is made. What are the possible outcomes, together with their probabilities, of the measurement? What is the probability for the system to be found in the excited state in a subsequent measurement of energy? **[8]**

(e) If the value of B measured at $t = 0$ was 1, what is the probability that a subsequent measurement will return the same value of B at a later time t? **[9]**

2. (a) What is meant by the term ***spin***? What values of spin, in units of $\hbar$, are allowed for fermions and bosons? **[2]**

(b) The normalised eigenkets of operator $\hat{S}_z$ for spin $\frac{1}{2}$ are given in matrix representation by $|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. By acting on these eigenkets by the operators $\hat{S}_\pm \equiv \hat{S}_x \pm i\hat{S}_y$, explain why they are called 'raising' and 'lowering' operators. **[6]**

(c) In a certain spin state the expectation value $\langle \hat{S}_x \rangle = \frac{3}{10}\hbar$. Show that the expectation value of the spin in a direction perpendicular to x, given by $\left\langle \hat{S}_\perp \right\rangle = \sqrt{\left\langle \hat{S}_y \right\rangle^2 + \left\langle \hat{S}_z \right\rangle^2}$, is equal to $\pm\frac{2}{5}\hbar$. **[6]**

[*Hint: You might find it useful to introduce a direction $\boldsymbol{n}$ in which the spin projection is maximal.*]

A system consisting of two spin-$\frac{1}{2}$ particles is in the eigenstate $|S, M\rangle = |1, 1\rangle$ of the total spin $\hat{\boldsymbol{S}}_{\text{tot}}$ and its z projection $\hat{S}_z$, with the eigenvalues $S = \hbar$ and $M = \hbar$. Let $|\sigma\rangle_i$ (with $\sigma = \pm\,,\ i = 1, 2$) denote the eigenkets corresponding to the eigenvalues $S_z = \pm\frac{1}{2}\hbar$ for the individual particles.

(d) Express $|1, 1\rangle$ in terms of the appropriate components of $|\sigma\rangle_1$ and $|\sigma\rangle_2$. **[4]**

(e) The lowering operator for the z component of the total spin is the sum of the lowering operators for the two particles. By applying the lowering operator repeatedly to the eigenstate in (d), find expressions for two other eigenstates in terms of $|\sigma\rangle_1$ and $|\sigma\rangle_2$. What are the corresponding eigenvalues S and M in these states? **[6]**

You are not required to normalize these eigenstates.

(f) Explain why the fourth eigenstate, $|S_0,\, M_0\rangle$, must exist. What are its eigenvalues S_0 and M_0? Using the fact that $|S_0,\, M_0\rangle$ must be orthogonal to the states found in (d) and (e), or otherwise, find an expression for it in terms of $|\sigma\rangle_1$ and $|\sigma\rangle_2$. **[6]**

3. A Hamiltonian of a particle of mass m in a parabolic potential well (the quantum harmonic oscillator) can be represented as

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2\hat{x}^2}{2} \equiv \hbar\omega\left(\hat{N} + \tfrac{1}{2}\right) ,$$

where $\hat{N} \equiv \hat{a}^\dagger\hat{a}$ is the number operator and the ladder operators $\hat{a}$ and $\hat{a}^\dagger$ are mutually adjoint with $\hat{a}$ defined via the position and momentum operators $\hat{x}$ and $\hat{p}$ by

$$\hat{a} \equiv \frac{1}{\sqrt{2}}\left(\hat{X} + i\hat{P}\right), \qquad \hat{X} \equiv \frac{\hat{x}}{\ell}, \quad \hat{P} \equiv \frac{\hbar\hat{p}}{\ell}, \quad \ell \equiv \left(\frac{\hbar}{m\omega}\right)^{\frac{1}{2}} .$$

(a) Assuming that the eigenvalue equation for the number operator is $\hat{N}\,|n\rangle = n\,|n\rangle$ and using the commutation relations (see Formula Sheet)

i. prove that n is a non-negative number;

ii. show that $\hat{a}$ acts as the lowering operator, $\hat{a}\,|n\rangle = \sqrt{n}\,|n{-}1\rangle$, and hence prove that n is integer and write the energy eigenvalues ε_n. **[10]**

(b) Calculate the expectation value of the particle's position and momentum in a superposition state, $|\psi\rangle = \frac{1}{2}\,|0\rangle + \frac{i\sqrt{3}}{2}\,|1\rangle$. **[10]**

When relativistic effects are incorporated in the lowest order into the motion of a simple harmonic oscillator, its Hamiltonian is modified to become

$$\hat{H}_{\rm rel} = \tfrac{1}{2}\hbar\omega\left(\hat{P}^2 + \hat{X}^2 - \lambda\hat{P}^4\right), \quad \text{where} \quad \lambda \equiv \frac{\hbar\omega}{4mc^2} \ll 1 .$$

(c) Use perturbation theory to calculate the lowest order relativistic correction to the energy eigenvalues ε_n of the simple harmonic oscillator. Comment on the applicability of your results for large n. **[10]**

Formula Sheet

Resolution of unity (completeness)
$\hat{I} = \sum_n |\eta_n\rangle\langle\eta_n| = \int\limits_{-\infty}^{\infty} \mathrm{d}x\, |x\rangle\langle x|$

Orthonormality: $\langle\eta_n|\,\eta_m\rangle = \delta_{nm}$

Hermitian conjugation: if $|\varphi\rangle = \hat{A}\,|\psi\rangle$, then $\langle\varphi| = \langle\psi|\hat{A}^\dagger$.

Ket and bra relations to the wavefunction:
$\langle x|\,\psi\rangle = \psi(x)\,;\quad \langle\psi|\,x\rangle = \psi^*(x)$.

Heisenberg uncertainty principle
$\Delta A \Delta B \geqslant \frac{1}{2}\left|\left\langle\left[\hat{A},\hat{B}\right]\right\rangle\right|$,
$\Delta A \equiv \sqrt{\langle\hat{A}^2\rangle - \langle\hat{A}\rangle^2}$

Time evolution $|\psi(t)\rangle = \hat{U}(t)\,|\psi(0)\rangle$,
$\hat{U}(t) = \mathrm{e}^{-i\hat{H}t/\hbar}$

Evolution of operators in Heisenberg picture
$\hat{A}_{\mathrm{H}}(t) = \hat{U}^\dagger\hat{A}(0)\hat{U}(t)$.

Ladder operators for SHO, $\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2\hat{x}^2}{2}$:

$\hat{a} = \frac{1}{\sqrt{2}}\left(\frac{\hat{x}}{\ell} + \frac{i\ell}{\hbar}\hat{p}\right) \equiv \frac{1}{\sqrt{2}}\left(\hat{X} + i\hat{P}\right)$, $\ell \equiv \sqrt{\frac{\hbar}{m\omega}}$,

$\hat{X} = \frac{1}{\sqrt{2}}\left(\hat{a} + \hat{a}^\dagger\right),\ \hat{P} = \frac{i}{\sqrt{2}}\left(\hat{a}^\dagger - \hat{a}\right)$

Commutation relations for $\hat{N} \equiv \hat{a}^\dagger\hat{a}$:
$[\hat{a},\hat{a}^\dagger] = 1\,,\quad \left[\hat{a},\hat{N}\right] = \hat{a}\,,\quad \left[\hat{a}^\dagger,\hat{N}\right] = -\hat{a}^\dagger$.

Action on eigenstates of $\hat{N}$:
$\hat{a}\,|n\rangle = \sqrt{n}\,|n{-}1\rangle,\ \hat{a}^\dagger\,|n\rangle = \sqrt{n{+}1}\,|n{+}1\rangle$.

Angular momentum commutation relations
$[\hat{L}_x,\hat{L}_y] = i\hbar L_z$, $[\hat{L}_y,\hat{L}_z] = i\hbar L_x$,
$[\hat{L}_z,\hat{L}_x] = i\hbar L_y$; $[\hat{L}_k,\hat{L}^2] = 0,\ k = x,y,z$.

Orbital angular momentum eigenstates
$\hat{L}^2\,|\ell,m\rangle = \hbar^2\ell(\ell+1)\,|\ell,m\rangle$,
$\hat{L}_z\,|\ell,m\rangle = \hbar m\,|\ell,m\rangle$
$\ell = 0,1,2,\ldots;\ m = 0,\pm1,\ldots,\pm\ell$.

Ladder operators $\hat{L}_\pm = \hat{L}_x \pm i\hat{L}_y$,
$[\hat{L}_+,\hat{L}_-] = 2\hbar\hat{L}_z$, $[\hat{L}_z,\hat{L}_\pm] = \pm\hbar\hat{L}_\pm$
$\hat{L}_\pm\,|\ell,m\rangle = \hbar c^\pm_{\ell,m}\,|\ell,m\pm1\rangle$
$c^\pm_{\ell,m} = \sqrt{(\ell\mp m)(\ell\pm m+1)}$

Spin has the same properties with $\hat{L}_i \to \hat{S}_i$, $\ell \to s$, which is integer or half-integer

Spin $\frac{1}{2}$ with $\hat{S}_k = \frac{1}{2}\hbar\hat{\sigma}_k$ where Pauli matrices

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},\ \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$
$$\hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Dirac notations for spin $\frac{1}{2}$:
$|\frac{1}{2},\frac{1}{2}\rangle \equiv |{+}\rangle \equiv |{\uparrow}\rangle,\ |\frac{1}{2},-\frac{1}{2}\rangle \equiv |{-}\rangle \equiv |{\downarrow}\rangle$.

Time-independent perturbation theory
$(\hat{H}_0 + \hat{V})\,|\psi_n\rangle = E_n\,|\psi_n\rangle$, $\hat{H}_0\,|n\rangle = E_n^{(0)}\,|n\rangle$
$E_n \simeq E_n^{(0)} + E_n^{(1)} + E_n^{(2)}$
$E_n^{(1)} = \langle n|\hat{V}\,|n\rangle,\ E_n^{(2)} = \sum_{m\neq n}\frac{|\langle m|\hat{V}\,|n\rangle|^2}{E_n^{(0)} - E_m^{(0)}}$
$|\psi_n\rangle \simeq |n\rangle + \sum_{m\neq n}\frac{\langle m|\hat{V}\,|n\rangle}{E_n^{(0)} - E_m^{(0)}}\,|m\rangle$

Time-dependent perturbation theory.
Transition probability $|i\rangle \to |f\rangle$
$P_{i\to f} = |d_{fi}(t)|^2$,
$d_{fi}(t) = \delta_{fi} - \frac{i}{\hbar}\int_{t_0}^{t}\mathrm{d}t'\langle f|\hat{V}(t')\,|i\rangle\mathrm{e}^{it'\omega_{fi}}$,
$\omega_{fi} = \frac{E_f - E_i}{\hbar}$.

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.